

AMARNATH REDDY MALKIREDDY

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PROFESSIONAL SUMMARY

- Analytics professional with 6+ years of experience turning complex healthcare, financial, academic, and operational data into clean datasets, reliable reports, KPI dashboards, and business-ready insights using SQL, Python, Power BI, Tableau, BigQuery, AWS, and Snowflake.
- Strong hands-on background in data cleaning, data pipelines, source-to-target reconciliation, missing-value handling, data quality checks, coverage analysis, schema validation, query optimization, and recurring reporting workflows.
- Experienced working with analysts, business users, and policy stakeholders to clarify requirements, define metrics, document logic, and deliver trusted reporting outputs that improve visibility, audit readiness, and decision-making.

CORE SKILLS

Data Analysis: KPI analysis, trend analysis, variance analysis, root-cause analysis, exploratory data analysis, data profiling, operational analytics, compliance analytics, financial transaction analysis, stakeholder reporting

SQL & Python: Advanced SQL, Python, Pandas, NumPy, CTEs, joins, window functions, query optimization, data cleaning, data transformation, reusable scripts, JSON/CSV/fixed-width data processing

Data Quality & Reconciliation: Source-to-target validation, missing-value handling, null checks, type validation, duplicate detection, schema validation, coverage checks, exception reporting, file-level sanity checks, data acceptance testing

Data Pipelines & Platforms: ETL/ELT workflows, batch ingestion, incremental loads, MERGE/UPSERT logic, BigQuery, GCS, GCP, AWS S3, AWS Glue, AWS Redshift, Snowflake, dbt, Airflow, Cloud Composer

BI & Reporting: Power BI, Tableau, Excel, Power Query, PivotTables, XLOOKUP, dashboard development, KPI reporting, recurring report automation, regulatory reporting, executive-ready summaries

Workflow & Documentation: GitHub, GitHub Actions, REST APIs, production monitoring, issue logs, methodology documentation, stakeholder communication, repeatable reporting processes

PROFESSIONAL EXPERIENCE

New York State Department of Health

Bioinformatics Data Engineer

Albany, NY

02/2024 – Present

- Built and maintained BigQuery ETL/ELT pipelines and GCS-to-BigQuery ingestion workflows for IQIES and ASPEN healthcare datasets, processing 1M+ records and supporting 8+ recurring statewide operational, staffing, census, provider, CMS, and compliance reports.
- Turned raw healthcare and regulatory data into validated reporting tables, KPI datasets, and reusable analytical layers used by analysts and program teams for monitoring, planning, and operational reviews.
- Created SQL and Python validation workflows across 1,500+ fields for SAS-to-BigQuery MDS comparisons, catching schema drift, type mismatches, null differences, whitespace issues, and record-level discrepancies before release.
- Improved report refresh and query performance by 45% by redesigning SQL transformations, Python preprocessing steps, incremental MERGE logic, partitioning, clustering, and BigQuery workload patterns.
- Built data quality checks for duplicate records, missing values, null/type issues, malformed files, coverage gaps, and file-level sanity testing across monthly and quarterly healthcare feeds.
- Standardized provider, facility, census, staffing, and program-level data structures so recurring reports used consistent logic across overlapping reporting needs.
- Partnered with analysts, program teams, and policy stakeholders to clarify reporting questions, define KPIs, document business rules, and deliver clean datasets and dashboards for decision support.

University at Albany (SUNY)

Data Engineer, Graduate Assistant

Albany, NY

04/2022 – 05/2023

- Standardized 100K+ academic records using SQL and Python workflows, improving reporting consistency and reducing manual data preparation effort by 40%.
- Built reusable validation routines and clean analytical tables for missing-value handling, schema consistency, source-field normalization, and downstream reporting.
- Prepared analysis-ready datasets by cleaning raw fields, applying consistent schemas, documenting transformation logic,

and creating repeatable data preparation workflows.

Accenture
Data Engineer

Hyderabad, India
01/2019 – 12/2021

- Built AWS and Snowflake ELT pipelines using Python, SQL, AWS Glue, S3, Redshift, and Snowflake to process \$20M+ in daily financial transactions for fraud analytics, AML/KYC monitoring, risk operations, and regulatory reporting.
- Developed customer, account, transaction, and entity-level datasets used for fraud monitoring, suspicious-activity review, customer segmentation, investigator prioritization, and recurring compliance reporting.
- Created source-to-target reconciliation, null validation, duplicate checks, coverage checks, and missing-value handling routines to improve completeness, accuracy, and audit readiness across regulated financial datasets.
- Designed SQL transformation logic across customer, account, transaction, product, and entity-level data to support consistent reporting for business, risk, and compliance teams.
- Used trend analysis, anomaly investigation, segmentation logic, and SQL-based pattern discovery to identify suspicious transaction behavior and support faster case triage for banking operations teams.
- Reduced batch runtime by 80% by redesigning ETL components, optimizing SQL logic, tuning warehouse workloads, and improving processing efficiency for large-volume financial datasets.
- Developed Power BI and Tableau dashboards for operational KPIs, exception tracking, fraud/AML monitoring, Basel-aligned risk reporting, and compliance metrics.
- Automated production monitoring and reporting checks with CI/CD and cloud-native logging, improving SLA adherence and reducing downstream rework during month-end and regulatory reporting cycles.

PROJECTS

Data Quality Guardian

Python, Pandas, NumPy, Streamlit

- Built a Streamlit-based data quality application that checks incoming datasets for missing values, duplicate records, schema drift, row-level issues, and statistical outliers before downstream reporting or analysis.
- Generated exportable issue logs, visual summaries, and reusable preprocessing outputs to make data-quality review faster, easier to explain, and more repeatable.
- Added configurable validation checks and row-level quality outputs to support cleaner dataset preparation, coverage review, root-cause analysis, and acceptance testing.

Job Alerts Automation

Python, REST APIs, GitHub Actions, SendGrid

- Developed an automated Python workflow that pulls job postings from public ATS REST APIs, normalizes semi-structured JSON responses into a common schema, removes duplicates, ranks relevant postings, and sends hourly HTML email digests.
- Added scoring logic, caching, monitoring, and failure alerts to make the workflow reliable, repeatable, and easier to maintain.
- Standardized API payloads for downstream filtering, ranking, delivery, and reusable analytics workflows.

EDUCATION

University at Albany (SUNY) — Albany, NY
Master of Science in Data Science

12/2023

Academic focus: Machine learning, statistical modeling, predictive analytics, and NLP fundamentals

GITAM University — India
Bachelor of Technology in Electronics and Communication Engineering

04/2019